

| Function                | CC                                |                            |                                   |                                   | NDAs                              |                            |                            |                                   | HCC                               | Ours                          |
|-------------------------|-----------------------------------|----------------------------|-----------------------------------|-----------------------------------|-----------------------------------|----------------------------|----------------------------|-----------------------------------|-----------------------------------|-------------------------------|
|                         | DG2-CMAES                         | Random-CMAES               | RDG3-CMAES                        | RDDSM-CMAES                       | Sep-CMAES                         | LM-MA-ES                   | LMCMA                      | MM-ES                             | HCC-ES                            | ARAC                          |
| E1                      | 3.15E+06 (+)<br>± 2.14E+05        | 9.80E+11 (+)<br>± 1.53E+11 | 3.13E+06 (+)<br>± 2.43E+05        | 2.88E+06 (+)<br>± 3.12E+05        | 5.62E+08 (+)<br>± 4.51E+04        | 3.81E+08 (+)<br>± 6.44E+04 | 8.69E+07 (+)<br>± 3.43E+03 | 1.60E+07 (+)<br>± 4.21E+03        | 2.84E+06 (+)<br>± 2.35E+05        | <b>5.69E+05</b><br>± 1.57E+06 |
| E2                      | 2.27E+11 (+)<br>± 1.31E+11        | 3.56E+08 (+)<br>± 1.01E+11 | 6.72E+08 (+)<br>± 4.15E+06        | 6.34E+08 (+)<br>± 5.24E+06        | 5.43E+08 (+)<br>± 6.23E+04        | 3.40E+08 (+)<br>± 2.39E+04 | 9.54E+07 (+)<br>± 2.77E+03 | 1.77E+07 (+)<br>± 5.78E+03        | 6.87E+06 (+)<br>± 4.73E+04        | <b>5.62E+06</b><br>± 3.78E+06 |
| E3                      | 2.38E+11 (+)<br>± 2.03E+11        | 5.45E+11 (+)<br>± 2.78E+11 | 2.02E+08 (+)<br>± 8.46E+06        | 2.07E+08 (+)<br>± 2.53E+06        | 6.97E+08 (+)<br>± 6.84E+04        | 4.32E+08 (+)<br>± 3.42E+04 | 1.40E+08 (+)<br>± 9.37E+04 | 1.99E+07 (+)<br>± 7.55E+03        | 1.60E+07 (+)<br>± 5.62E+05        | <b>1.34E+07</b><br>± 5.20E+06 |
| E4                      | 2.70E+11 (+)<br>± 7.93E+10        | 4.75E+09 (+)<br>± 3.44E+11 | 6.35E+09 (+)<br>± 5.10E+06        | 4.86E+09 (+)<br>± 4.81E+07        | 4.89E+08 (+)<br>± 7.23E+04        | 4.89E+08 (+)<br>± 5.53E+04 | 1.13E+08 (+)<br>± 2.77E+04 | <b>1.90E+07 (-)</b><br>± 5.45E+03 | 2.26E+07 (-)<br>± 3.04E+05        | 2.61E+07<br>± 9.35E+06        |
| E5                      | 2.86E+11 (+)<br>± 1.75E+11        | 7.23E+11 (+)<br>± 2.53E+11 | 1.29E+10 (+)<br>± 4.11E+07        | 1.10E+10 (+)<br>± 6.23E+07        | 7.45E+08 (+)<br>± 5.31E+04        | 4.61E+08 (+)<br>± 2.31E+04 | 1.63E+08 (+)<br>± 3.09E+04 | 2.48E+07 (-)<br>± 4.29E+03        | <b>7.76E+06 (-)</b><br>± 2.13E+05 | 2.98E+07<br>± 9.18E+06        |
| E6                      | 3.20E+11 (+)<br>± 9.22E+10        | 5.19E+11 (+)<br>± 1.21E+11 | 3.88E+10 (+)<br>± 3.22E+07        | 3.42E+10 (+)<br>± 1.69E+08        | 7.92E+08 (+)<br>± 3.07E+04        | 4.71E+08 (+)<br>± 4.02E+04 | 1.44E+08 (+)<br>± 4.14E+04 | <b>2.62E+07 (-)</b><br>± 2.37E+03 | 4.32E+07 (+)<br>± 1.94E+05        | 3.19E+07<br>± 6.54E+06        |
| S1                      | 1.98E-03 (-)<br>± 5.18E-04        | 9.67E+08 (+)<br>± 1.13E+08 | <b>1.37E-03 (-)</b><br>± 4.23E-04 | 1.46E-03 (-)<br>± 5.29E-04        | 5.06E+06 (+)<br>± 1.66E+02        | 2.22E+05 (+)<br>± 3.51E+01 | 5.50E+05 (+)<br>± 3.05E+01 | 1.67E+04 (+)<br>± 2.14E+01        | 1.92E-03 (-)<br>± 3.82E-04        | 1.04E+00<br>± 5.12E+00        |
| S2                      | 4.67E+08 (+)<br>± 5.28E+07        | 1.04E+09 (+)<br>± 5.26E+08 | 7.92E+05 (+)<br>± 5.26E+04        | 8.10E+04 (+)<br>± 5.43E+03        | 2.63E+06 (+)<br>± 2.15E+02        | 2.20E+05 (+)<br>± 1.41E+01 | 7.76E+05 (+)<br>± 2.23E+01 | 1.30E+04 (+)<br>± 7.54E+01        | 5.58E+03 (+)<br>± 3.21E+01        | <b>8.12E+02</b><br>± 5.82E+02 |
| S3                      | 4.57E+08 (+)<br>± 3.08E+07        | 9.97E+08 (+)<br>± 1.21E+08 | 7.60E+06 (+)<br>± 6.13E+05        | 2.48E+06 (+)<br>± 4.60E+05        | 4.14E+06 (+)<br>± 3.43E+02        | 2.43E+05 (+)<br>± 3.12E+01 | 4.89E+05 (+)<br>± 5.13E+01 | 1.91E+04 (+)<br>± 9.47E+01        | 9.72E+03 (+)<br>± 4.75E+01        | <b>2.46E+03</b><br>± 1.35E+03 |
| S4                      | 5.43E+08 (+)<br>± 2.84E+07        | 8.44E+08 (+)<br>± 2.37E+08 | 2.03E+08 (+)<br>± 2.37E+07        | 2.88E+07 (+)<br>± 9.25E+06        | 3.14E+06 (+)<br>± 5.67E+02        | 2.54E+05 (+)<br>± 4.27E+01 | 5.35E+05 (+)<br>± 7.42E+01 | 1.50E+04 (+)<br>± 1.05E+02        | <b>1.24E+03 (-)</b><br>± 5.31E+01 | 5.11E+03<br>± 3.49E+03        |
| S5                      | 6.83E+08 (+)<br>± 7.42E+07        | 1.32E+09 (+)<br>± 5.91E+08 | 8.84E+07 (+)<br>± 3.54E+06        | 6.77E+06 (+)<br>± 4.60E+05        | 2.14E+06 (+)<br>± 3.48E+02        | 3.01E+05 (+)<br>± 3.65E+01 | 1.32E+06 (+)<br>± 9.13E+01 | 1.87E+04 (+)<br>± 8.53E+01        | 9.23E+03 (+)<br>± 3.95E+02        | <b>8.38E+03</b><br>± 8.69E+03 |
| S6                      | 7.01E+08 (+)<br>± 6.12E+07        | 1.06E+09 (+)<br>± 3.13E+08 | 9.01E+07 (+)<br>± 3.13E+06        | 6.01E+06 (+)<br>± 5.73E+05        | 2.86E+06 (+)<br>± 5.73E+02        | 2.26E+05 (+)<br>± 4.42E+01 | 7.38E+05 (+)<br>± 7.82E+01 | 1.33E+04 (+)<br>± 7.19E+01        | 6.65E+04 (+)<br>± 7.36E+02        | <b>4.18E+03</b><br>± 1.99E+03 |
| R1                      | 1.87E+05 (+)<br>± 3.11E+03        | 4.07E+08 (+)<br>± 1.13E+08 | 1.74E+05 (+)<br>± 6.67E+03        | 1.76E+05 (+)<br>± 2.03E+03        | 2.03E+05 (+)<br>± 2.27E+01        | 7.92E+05 (+)<br>± 9.23E+01 | 3.55E+06 (+)<br>± 3.63E+02 | 1.36E+06 (+)<br>± 4.01E+02        | 1.74E+05 (+)<br>± 8.13E+03        | <b>1.56E+05</b><br>± 1.53E+04 |
| R2                      | 4.39E+07 (+)<br>± 2.04E+06        | 7.96E+08 (+)<br>± 2.25E+08 | 1.47E+06 (+)<br>± 2.83E+05        | 3.02E+05 (+)<br>± 1.37E+04        | 2.48E+05 (+)<br>± 3.82E+01        | 8.30E+05 (+)<br>± 8.39E+01 | 2.17E+06 (+)<br>± 2.31E+02 | 1.39E+06 (+)<br>± 3.71E+02        | 3.72E+05 (+)<br>± 5.27E+03        | <b>2.24E+05</b><br>± 2.95E+04 |
| R3                      | 4.43E+07 (+)<br>± 6.83E+06        | 1.06E+09 (+)<br>± 4.77E+08 | 9.57E+05 (+)<br>± 9.89E+04        | 6.16E+05 (+)<br>± 2.64E+04        | 3.28E+05 (+)<br>± 4.13E+01        | 8.85E+05 (+)<br>± 6.55E+01 | 2.97E+06 (+)<br>± 2.52E+02 | 1.34E+06 (+)<br>± 6.57E+02        | 5.06E+05 (+)<br>± 7.26E+03        | <b>2.69E+05</b><br>± 3.29E+04 |
| R4                      | 4.61E+07 (+)<br>± 2.33E+06        | 1.39E+09 (+)<br>± 2.25E+08 | 2.60E+06 (+)<br>± 2.83E+05        | 7.93E+05 (+)<br>± 1.37E+04        | 3.20E+05 (+)<br>± 3.37E+01        | 8.97E+05 (+)<br>± 4.23E+01 | 2.36E+06 (+)<br>± 3.53E+02 | 1.48E+06 (+)<br>± 2.68E+02        | 5.59E+05 (+)<br>± 6.77E+03        | <b>3.15E+05</b><br>± 3.25E+04 |
| R5                      | 5.25E+07 (+)<br>± 4.11E+06        | 6.84E+08 (+)<br>± 1.56E+08 | 2.05E+06 (+)<br>± 3.21E+05        | 1.25E+06 (+)<br>± 8.98E+04        | <b>2.48E+05 (-)</b><br>± 1.92E+01 | 9.50E+05 (+)<br>± 2.19E+01 | 4.05E+06 (+)<br>± 2.84E+02 | 1.52E+06 (+)<br>± 3.55E+02        | 6.84E+05 (+)<br>± 5.76E+03        | 3.34E+05<br>± 4.10E+04        |
| R6                      | 5.86E+07 (+)<br>± 3.50E+06        | 1.41E+09 (+)<br>± 4.26E+08 | 3.85E+06 (+)<br>± 2.57E+05        | 1.30E+06 (+)<br>± 3.47E+05        | <b>2.66E+05 (-)</b><br>± 4.35E+01 | 9.57E+05 (+)<br>± 3.53E+01 | 3.03E+06 (+)<br>± 3.37E+02 | 1.68E+06 (+)<br>± 1.16E+02        | 8.15E+05 (+)<br>± 2.30E+04        | 3.61E+05<br>± 4.19E+04        |
| A1                      | <b>7.76E+04 (-)</b><br>± 1.03E+02 | 7.93E+04 (+)<br>± 1.13E+02 | <b>7.76E+04 (-)</b><br>± 9.04E+02 | <b>7.76E+04 (-)</b><br>± 1.07E+01 | 7.81E+04 (+)<br>± 1.76E+00        | 7.84E+04 (+)<br>± 7.31E+00 | 7.94E+04 (+)<br>± 1.79E+02 | 7.94E+04 (+)<br>± 2.78E+02        | <b>7.76E+04 (-)</b><br>± 1.25E+01 | 7.80E+04<br>± 9.33E+01        |
| A2                      | 8.09E+04 (+)<br>± 7.90E+02        | 7.97E+04 (+)<br>± 1.25E+02 | 8.07E+04 (+)<br>± 8.93E+02        | 7.82E+04 (+)<br>± 2.33E+01        | 7.81E+04 (+)<br>± 2.44E+00        | 7.86E+04 (+)<br>± 1.34E+02 | 7.86E+04 (+)<br>± 2.55E+01 | 7.83E+04 (+)<br>± 9.32E+00        | 7.81E+04 (+)<br>± 9.78E+00        | <b>7.80E+04</b><br>± 9.27E+01 |
| A3                      | 8.10E+04 (+)<br>± 8.23E+02        | 7.98E+04 (+)<br>± 1.88E+02 | 8.12E+04 (+)<br>± 9.83E+02        | 7.81E+04 (≈)<br>± 3.01E+01        | <b>7.80E+04 (-)</b><br>± 3.89E+00 | 7.84E+04 (+)<br>± 8.78E+01 | 7.92E+04 (+)<br>± 1.03E+02 | 7.85E+04 (+)<br>± 1.53E+01        | 7.86E+04 (+)<br>± 2.09E+01        | 7.81E+04<br>± 9.28E+01        |
| A4                      | 8.12E+04 (+)<br>± 1.22E+03        | 7.96E+04 (+)<br>± 1.79E+02 | 7.90E+04 (+)<br>± 3.13E+00        | 7.83E+04 (+)<br>± 1.04E+01        | 7.84E+04 (+)<br>± 4.27E+01        | 7.85E+04 (+)<br>± 6.90E+01 | 7.87E+04 (+)<br>± 2.63E+01 | 7.89E+04 (+)<br>± 2.54E+02        | 7.83E+04 (+)<br>± 1.14E+01        | <b>7.82E+04</b><br>± 1.07E+02 |
| A5                      | 8.14E+04 (+)<br>± 1.39E+03        | 8.00E+04 (+)<br>± 1.93E+02 | 7.99E+04 (+)<br>± 3.04E+01        | 7.85E+04 (+)<br>± 3.15E+01        | <b>7.82E+04 (-)</b><br>± 3.42E+00 | 7.84E+04 (≈)<br>± 4.59E+02 | 8.06E+04 (+)<br>± 4.59E+02 | 7.87E+04 (+)<br>± 3.17E+01        | 7.85E+04 (+)<br>± 2.35E+01        | 7.84E+04<br>± 6.63E+01        |
| A6                      | 8.13E+04 (+)<br>± 1.12E+03        | 7.96E+04 (+)<br>± 1.57E+02 | 7.91E+04 (+)<br>± 2.11E+00        | 7.82E+04 (+)<br>± 5.23E+00        | <b>7.80E+04 (≈)</b><br>± 7.25E+00 | 7.84E+04 (+)<br>± 3.28E+01 | 7.99E+04 (+)<br>± 6.97E+02 | 7.89E+04 (+)<br>± 4.79E+01        | <b>7.80E+04 (≈)</b><br>± 7.06E+00 | 7.80E+04<br>± 7.73E+01        |
| ALL (+/≈/-)<br>vs. ARAC | 22/0/2                            | 24/0/0                     | 22/0/2                            | 21/1/2                            | 19/1/4                            | 23/1/0                     | 24/0/0                     | 21/0/3                            | 18/1/5                            | NA                            |